

Introduction to Biophysics II: Quantum physics & biology

Lecture 18: Quantum coherence in biology - 3

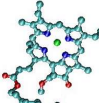
Reading:
Modern optical spectroscopy, chapter 10

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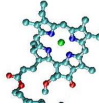
Last time....

Coherence: oscillations

Donor



Acceptor



Hamiltonian: $\tilde{H} = \tilde{H}_1 + \tilde{H}_2 + \tilde{H}_{21}$
interaction

Supermolecular state:

Excitation on donor

↓

Excitation on acceptor

↓

$$\Psi_B = C_1\psi_1 + C_2\psi_2 = C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b}$$

Solve time-dependent SE: $\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}$

Solution for $C_1(0)=1, C_2(0)=0$:

Population dynamics:

$|C_2(t)|^2 = (1/2)(H_{21}/\Omega)^2 [1 - \cos(2\Omega t/\hbar)]$

$$\Omega = (1/2) \left[(E_{21})^2 + 4(H_{21})^2 \right]^{1/2}$$

- Population on acceptor oscillates with period $T = 2\pi\hbar/2\Omega = \hbar/2\Omega$
- Mean probability is $(H_{21}/\Omega)^2/2$

$E_{21} = H_{22} - H_{11}$

Note: Ω is exactly the same as energy splitting in excitonic states (Lecture 15): $\pm \frac{1}{2} \sqrt{\delta^2 + 4(H_{12})^2}$

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Last time...

The density matrix

More generally, for any system that can be described by a linear combination of basis states, the expectation value A :

$$\Psi(t) = c_1(t)\psi_1 + c_2(t)\psi_2 + \dots$$

$$c_n(t) = C_n(t)\exp[-i(H_{nn}t/\hbar + \zeta_n)]$$

$$\langle A(t) \rangle = \sum_m \sum_n c_m^*(t) c_n(t) \langle \psi_m | \tilde{A} | \psi_n \rangle = \sum_m \sum_n c_m^*(t) c_n(t) A_{mn}$$

Basis *spatial* wavefunctions

All the observable time dependence of the system thus resides in the array of products of the coefficients

$$c_m^* c_n$$

Introduce

density matrix:

$$\rho_{nm}(t) = c_n(t) c_m^*(t)$$

This is product of two matrices
= $(\rho \mathbf{A})_{nn}$

Rewrite:

$$\langle A(t) \rangle = \sum_n \sum_m c_n(t) c_m^*(t) A_{mn} = \sum_n \left[\sum_m \rho_{nm}(t) A_{mn} \right]$$

$$\langle A(t) \rangle = \sum_n (\rho \mathbf{A})_{nn} = \text{Tr}(\rho \mathbf{A})$$

Tr $\equiv$ trace of a matrix,
sum of all diagonal elements

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Last time...

Von Neumann equation

$$\frac{\partial \rho}{\partial t} = (i/\hbar) [\rho, \mathbf{H}]$$

$$\rho_{nm}(t) = c_n(t) c_m^*(t)$$

$$\Psi(t) = \sum_n c_n(t) \psi_n(\mathbf{r})$$

$$c_n(t) = C_n(t) e^{-iH_{nn}t/\hbar}$$

Example:

Three-state system:

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{pmatrix} = \frac{i}{\hbar} \left[\begin{pmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix} - \begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix} \begin{pmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{pmatrix} \right]$$

$$\partial \rho_{11} / \partial t = (i/\hbar) (\rho_{11} H_{11} + \rho_{12} H_{21} + \rho_{13} H_{31} - \rho_{11} H_{11} - \rho_{21} H_{12} - \rho_{31} H_{13})$$

$$= (i/\hbar) (\rho_{12} H_{21} - \rho_{21} H_{12} + \rho_{13} H_{31} - \rho_{31} H_{13}).$$

What does $\rho_{11}(t)$ represent?

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Last time...

Von Neumann equation

Example: Three-state system:

$$\rho_{nm}(t) = c_n(t)c_m^*(t)$$

$$\begin{aligned} \text{populations} \quad & \begin{cases} \partial \rho_{11} / \partial t = (i/\hbar) (\rho_{12} H_{21} - \rho_{21} H_{12} + \rho_{13} H_{31} - \rho_{31} H_{13}) \\ \partial \rho_{22} / \partial t = (i/\hbar) (\rho_{21} H_{12} - \rho_{12} H_{21} + \rho_{23} H_{32} - \rho_{32} H_{23}) \\ \partial \rho_{33} / \partial t = (i/\hbar) (\rho_{31} H_{13} - \rho_{13} H_{31} + \rho_{32} H_{23} - \rho_{23} H_{32}) \end{cases} \\ \text{Off-diagonal} \quad & \begin{cases} \partial \rho_{12} / \partial t = (i/\hbar) [(\rho_{22} - \rho_{11}) H_{21} + \rho_{21} (H_{11} - H_{22})] + \rho_{23} H_{31} - \rho_{31} H_{23} \\ \partial \rho_{21} / \partial t = (i/\hbar) [(\rho_{22} - \rho_{11}) H_{21} + \rho_{21} (H_{11} - H_{22})] + \rho_{23} H_{31} - \rho_{31} H_{23} \\ \partial \rho_{13} / \partial t = (i/\hbar) [(\rho_{11} - \rho_{33}) H_{13} + \rho_{13} (H_{33} - H_{11}) + \rho_{12} H_{23} - \rho_{23} H_{12}] \\ \partial \rho_{31} / \partial t = (i/\hbar) [(\rho_{33} - \rho_{11}) H_{31} + \rho_{31} (H_{11} - H_{33}) + \rho_{32} H_{21} - \rho_{21} H_{32}] \\ \partial \rho_{23} / \partial t = (i/\hbar) [(\rho_{22} - \rho_{33}) H_{23} + \rho_{23} (H_{33} - H_{22}) + \rho_{21} H_{13} - \rho_{13} H_{21}] \\ \partial \rho_{32} / \partial t = (i/\hbar) [(\rho_{33} - \rho_{22}) H_{32} + \rho_{32} (H_{22} - H_{33}) + \rho_{31} H_{12} - \rho_{12} H_{31}] \end{cases} \end{aligned}$$

How do populations evolve?

- Need off-diagonal elements to build-up populations! is not instantaneous
- Consistent with what we had for 2-state system (red boxes above):

$$\partial C_1 / \partial t = -(i/\hbar) H_{12} \exp(iE_{12}t/\hbar) C_2$$

$$\partial C_2 / \partial t = -(i/\hbar) H_{21} \exp(iE_{21}t/\hbar) C_1$$

but now it depends on off-diagonal elements, not on populations directly!

Note: populations are balanced: $\partial \bar{\rho}_{mm} / \partial t = - \sum_{n \neq m} \partial \bar{\rho}_{mn} / \partial t$

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Last time...

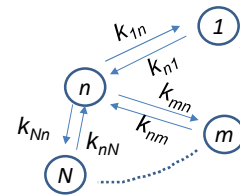
The Stochastic Liouville Equation

Assume interaction with surrounding:

$$\{\partial \bar{\rho}_{nn} / \partial t\}_{stochastic} = \sum_{m \neq n} (k_{nm} \bar{\rho}_{mm} - k_{mn} \bar{\rho}_{nn})$$

Notes:

- evolution above does not depend on the off-diagonal elements of $\bar{\rho}$
- due to Boltzmann equilibrium $\rightarrow \frac{k_{mn}}{k_{nm}} = \exp\left(-\frac{E_{nm}}{kT}\right)$



$$\text{Diagonal} \quad \left[\partial (\bar{\rho}_{nn} - \bar{\rho}_{nn}^0) / \partial t \right]_{stochastic} = -\frac{1}{T_1} (\bar{\rho}_{nn} - \bar{\rho}_{nn}^0)$$

T_1 is called *longitudinal relaxation time*

$$\text{Off-diagonal} \quad \{\partial \bar{\rho}_{nm} / \partial t\}_{stochastic} = -\frac{1}{T_2} \bar{\rho}_{nm} \quad \text{for } m \neq n, \text{ and } 1/T_2 \approx 1/T_2^* + 1/2T_1$$

T_2 – the *transverse relaxation time*, or just T2-time

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Last time...

The Stochastic Liouville Equation

Dynamics of diagonal and off-diagonal elements:

$$\partial \overline{\rho}_{nm} / \partial t = (i/\hbar) [\overline{\rho}, \mathbf{H}]_{nm} + \sum_{j,k} R_{nm,jk} \overline{\rho}_{jk}$$

↓

$$\{\partial \overline{\rho}_{nm} / \partial t\}_{\text{stochastic}} = -\frac{1}{T_2} \overline{\rho}_{nm}$$

Stochastic Liouville Equation:

$$\partial \overline{\rho} / \partial t = (i/\hbar) [\overline{\rho}, \mathbf{H}] + \mathbf{R} \overline{\rho}$$

von Neumann for system
+ Rρ for bath

$\mathbf{R}$ – relaxation matrix. = set of rate constants for stochastic processes.

$\mathbf{R}$ is in fact a fourth-order tensor, while ρ is second order tensor (or matrix).

$$(\mathbf{R} \overline{\rho})_{nm} = \sum_{i,k} R_{nmjk} \overline{\rho}_{jk}$$

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Last time....

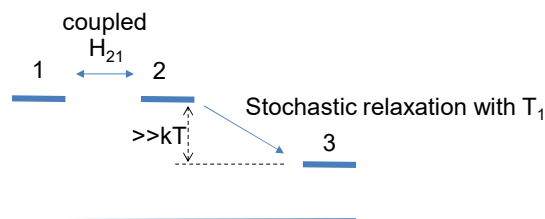
The Stochastic Effects

$$\partial \overline{\rho} / \partial t = (i/\hbar) [\overline{\rho}, \mathbf{H}] + \mathbf{R} \overline{\rho}$$

Example: 3 states:

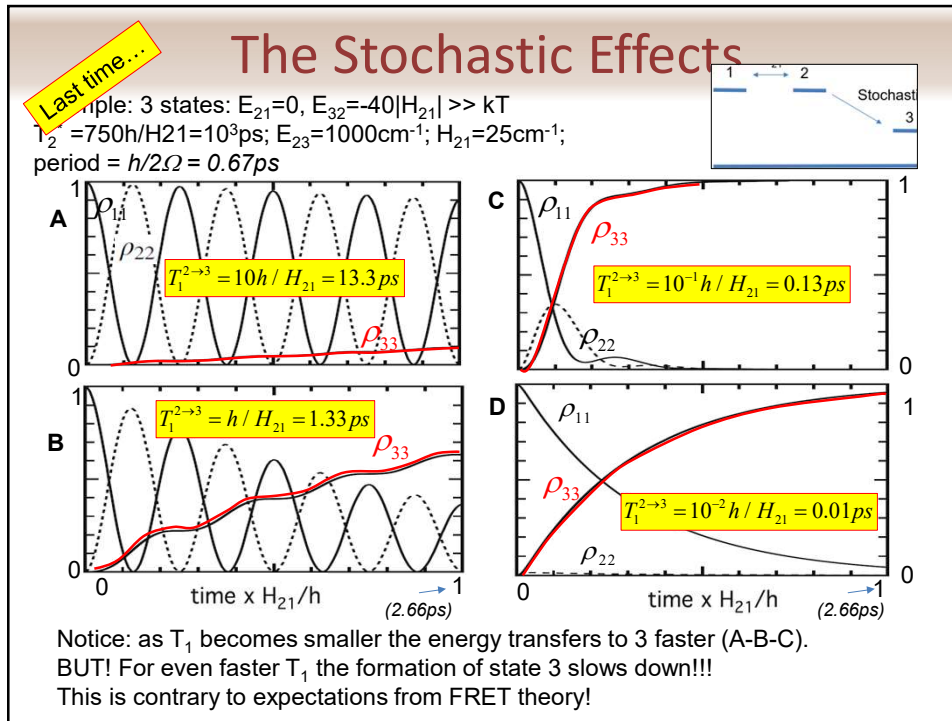
- States 1 & 2 are at the same energy ($E_{21}=0$) and interact ($H_{21} \neq 0$), interconversion is driven by QM coupling only
- State 2 can stochastically relax to state 3 at lower energy – resonance energy transfer
- Assume pure dephasing $T_2^* = 1 \text{ ns} \rightarrow$ is very long

i.e. $R_{11}=R_{22}=0$; $H_{13}=0$

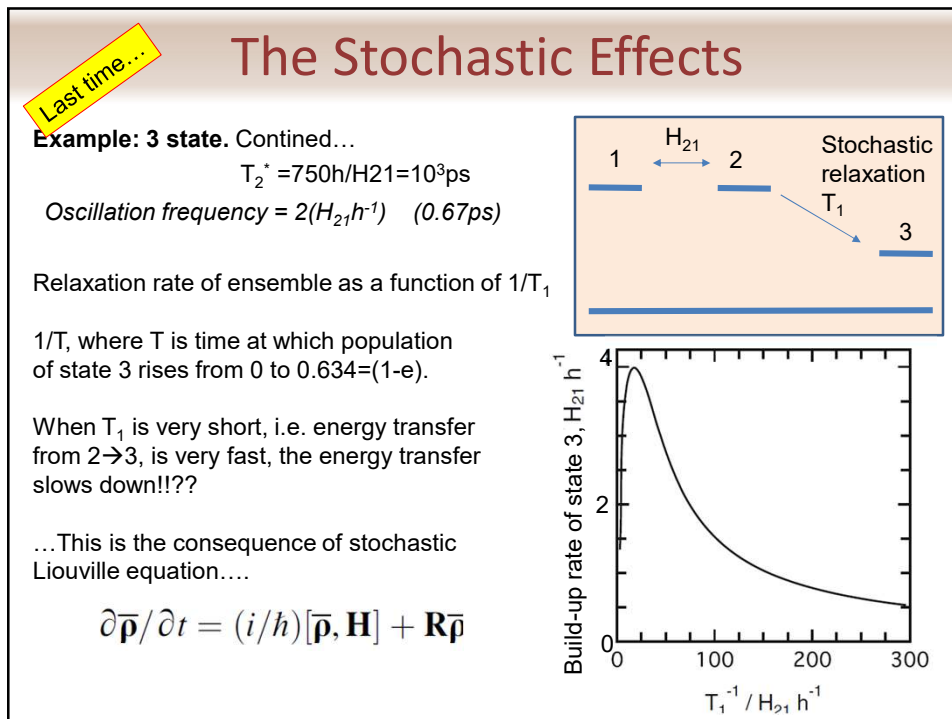


Ensemble starts in state 1
How rapidly will it appear in state 3?

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The Stochastic Effects

Example: 3 state. Continued...

When energy transfer from $2 \rightarrow 3$, is very fast, the energy transfer $1 \rightarrow 3$ slows down!??

1. Rate of transfer to (3) depends on:
 - Transfer rate $(2 \rightarrow 3)$, i.e. $1/T_1$
 - population of (2):

$$C_2(t) = -i(H_{21}/\Omega) \sin(\Omega t/\hbar) \exp(-iE_{21}t/2\hbar)$$

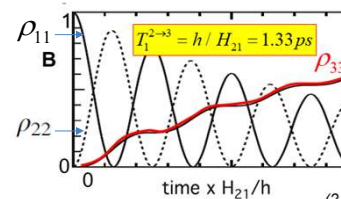
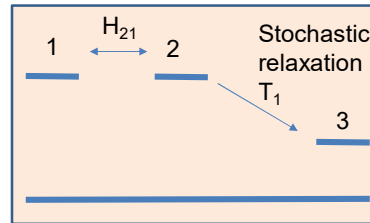
$$|C_2(t)|^2 = \left| \frac{H_{21}}{\Omega} \right|^2 \sin^2\left(\frac{\Omega t}{\hbar}\right)$$

2. Transfer $(2 \rightarrow 3)$ occurs at time $\sim T_1$, when population is:

$$|C_2(t)|^2 \propto \sin^2\left(\frac{\Omega}{\hbar} T_1\right) \propto T_1^2 \quad (\text{for small } T_1)$$

3. Rate of transfer to (3) is thus $\text{Rate}(3) = \text{Population}(2) \times 1/T_1 \propto T_1^2 \times T_1^{-1} = T_1$

i.e. rate of population (3) formation becomes *smaller* (slower!) when time T_1 becomes *smaller* (i.e. faster!)



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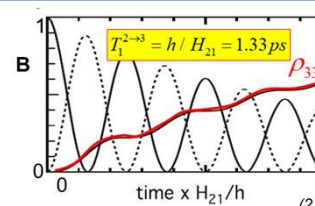
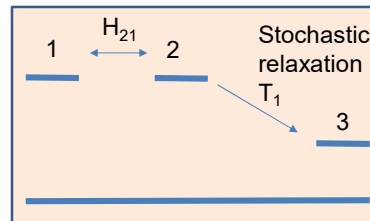
The Stochastic Effects

Example: 3 state. Continued...

When energy transfer from $2 \rightarrow 3$, is very fast, the energy transfer $1 \rightarrow 3$ slows down!??

In other words:

- Stochastic process $2 \rightarrow 3$ constantly tests the state of quantum system 1+2 with time interval $\sim T_1$.
- Each "test" brings the system to a *definite* state (1), (2) or (3).
- At short times probability to find system in state (2) is small ($\sim T_1^2$) and system is thus collapsed into definite state (1) and we are at "zero" time on the graph again!



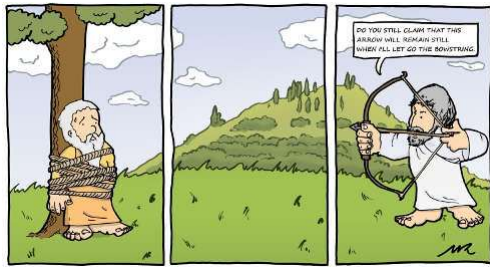
Quantum Zeno effect: particle's time evolution is arrested by measuring it frequently enough with respect to some chosen measurement setting

This is similar to the "quantum arrow" question – will it hit you faster when you look at it, or when you don't?

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Classical Zeno paradox

The name comes from Zeno's arrow paradox, which states that because an arrow in flight is not seen to move during any single instant, it cannot possibly be moving at all



(Somewhat similar to Achilles & turtle paradox)



Zeno of Elea
~490 BC – 430 BC

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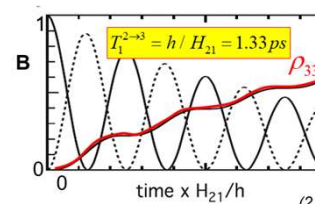
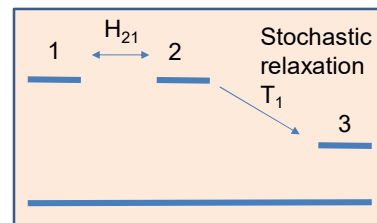
The Stochastic Effects

Example: 3 state. Continued...

When energy transfer from $2 \rightarrow 3$, is very fast, the energy transfer $1 \rightarrow 3$ slows down!!??

Alternative explanation:

- Rapid decay $2 \rightarrow 3$ shortens its lifetime and broadens its homogeneous energy distribution.
- According to Golden Fermi Rule the average rate depends on the density of states in the region $E_{12}=0$, which will thus decrease.



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Quantum Zeno effect

Quantum Zeno effect: particle's time evolution is arrested by measuring it frequently enough with respect to some chosen measurement setting

Each measurement causes the wavefunction to collapse to an eigenstate of the measurement basis.

Rigorous theory: 1974 (Degasperis, A.; Fonda, L.; Ghirardi, G. C.)
doi:10.1007/BF02731351

Experimental:

1989 David J. Wineland (NIST): 3-level Be ions in a spatial trap at low T

2001 Mark Raizen (University of Texas, Austin): ultracold sodium atoms)

2015 Mukund Vengalattore (Cornell Uni): ultracold lattice gas

See for more: https://en.wikipedia.org/wiki/Quantum_Zeno_effect

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Quantum Zeno effect

Experimental: 1989 Itano, Heinzen, Bollinger and Wineland (NIST)

Itano, W. M., Heinzen, D. J., Bollinger, J. J. and Wineland, D. J., Quantum Zeno effect. Phys. Rev. A, 1990, 41, 2295–2300

~5,000 $^9\text{Be}^+$ ions confined in EM trap and laser-cooled to below 250 mK

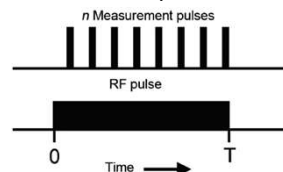
Levels S1 and S2 are stable excited states

Level S3 emits photon

1. Laser pulse: (ground) $\rightarrow$ S1

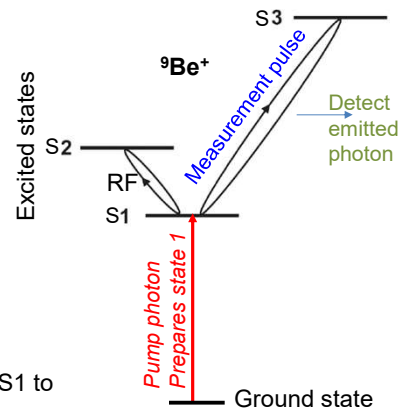
2. RF magnetic π pulse: S1 $\rightarrow$ S2

During the time that the RF pulse was applied, a variable number n of equally spaced short UV laser pulses were applied:



The measurement cycles system from state S1 to S3 and back to S1 collapsing wavefunction effectively to state S1.

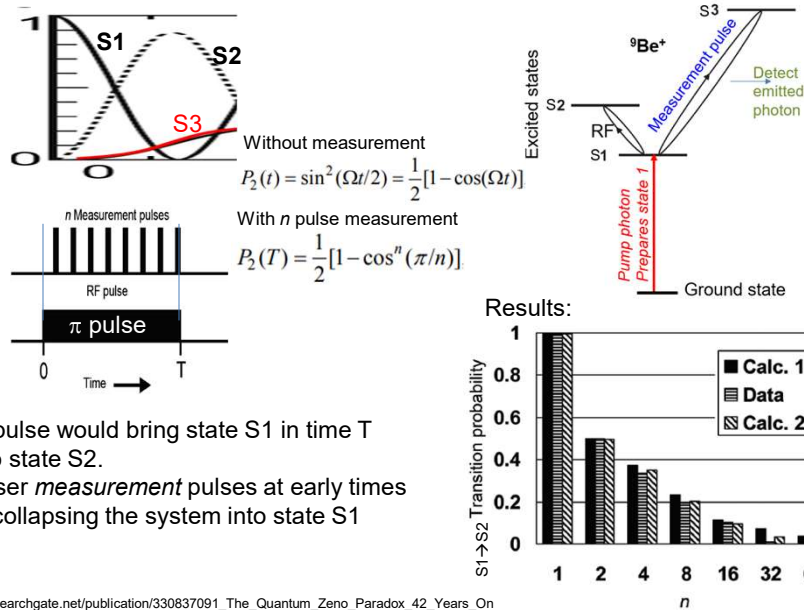
Emitted photon intensity corresponds to population of S1



https://www.researchgate.net/publication/330837091_The_Quantum_Zeno_Paradox_42_Years_On

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Quantum Zeno effect



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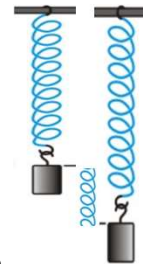
Classical analog: spring-mass oscillators

The stochastic Liouville – classic analogue
Recall two coupled spring-mass oscillators:

$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12} \quad \omega_i = \sqrt{k_i/m_i}$$

$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21} \quad \omega_{ci} = \sqrt{k_{12}/m_i}$$

Lets introduce T_2^* - phase noise due to some external random perturbation. In the program add a probability of phase shift at some random time, here I introduced phase noise by switching coordinate x_1 to $-x_1$ or x_2 to $-x_2$ at random times (this will keep energy constant) (can swap velocity direction instead)



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Classical analog: spring-mass oscillators

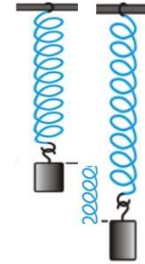
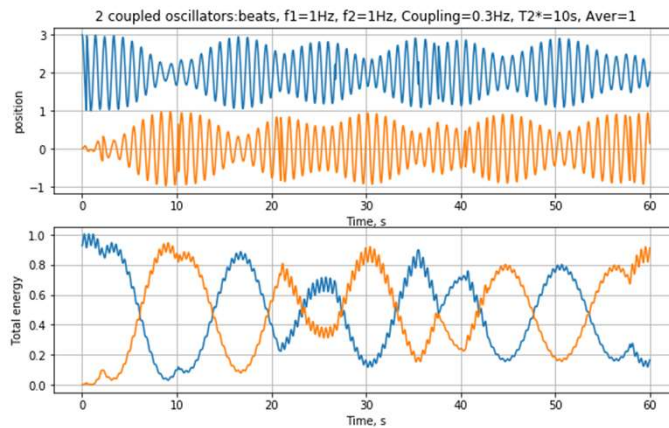
Simulation using simple Python script:

T=0: spring 1 (blue) is stretched, spring 2 (orange) is at equilibrium

Resonance frequencies: $\omega_1 = \omega_2$

Random flip of position $x \rightarrow -x$ with $T_2^* = 10$ s

Single trajectory!



$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$

$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

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Classical analog: spring-mass oscillators

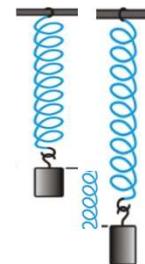
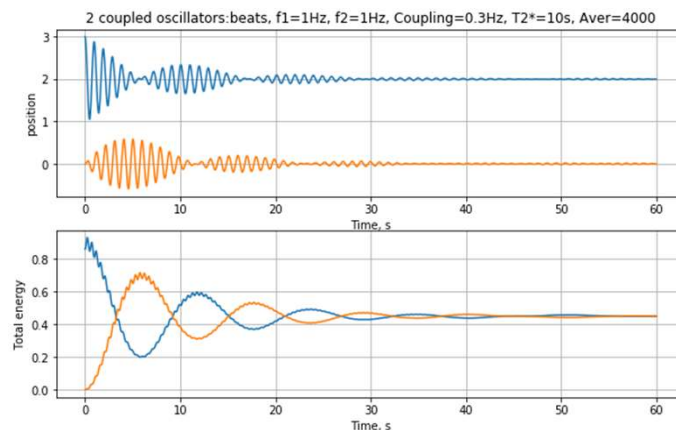
Simulation using simple Python script:

T=0: spring 1 (blue) is stretched, spring 2 (orange) is at equilibrium

Resonance frequencies: $\omega_1 = \omega_2$

Random flip of position $x \rightarrow -x$ with $T_2^* = 10$ s

Average of ensemble of 4000 systems

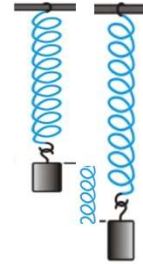
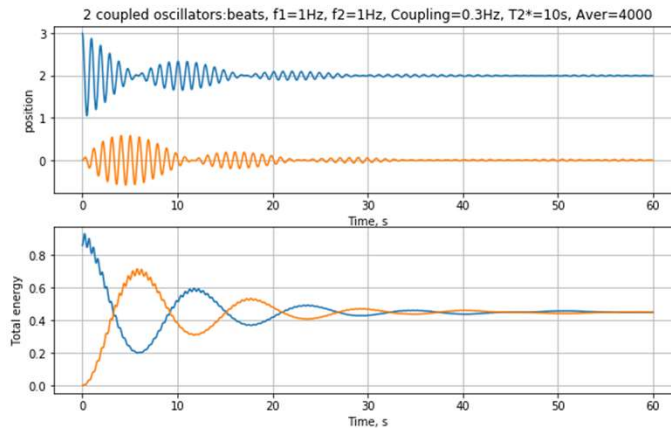


$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$

$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

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Classical analog: spring-mass oscillators



$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$

$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

Discuss:

- Can we observe Zeno effect in this classical system?
- What is it that makes quantum system different from classical?

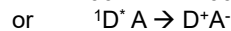
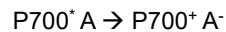
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The Stochastic Effects

Example: 3 state. Continued...

Similar scenario occurs in electron transfer in RC:

1) Electron transfer



Spins of unpaired electrons on the donor and acceptor remain initially antiparallel.

→ The radical pair is initially in a singlet state

2) Antiparallel electrons are on separate molecules

→ singlet and triplet states have similar energy

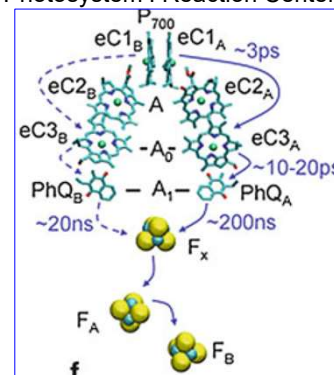
→ interaction of electron spin with nuclear spins can cause electron spin to oscillate between triplet and singlet states

3) If forward electron transfer is blocked, electrons can recombine readily from the triplet state to the triplet state of the donor since that has lower energy:



According to our analysis, the rate of ${}^3D^+$ formation will first increase with ${}^3(D^+ A^-) \rightarrow {}^3D^+ A$ rate, and then decrease.

Photosystem I Reaction Center



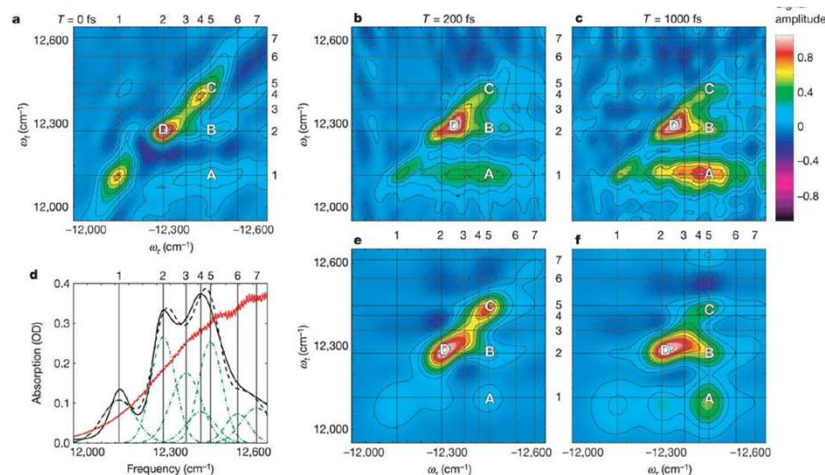
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Quantum coherence

Two-dimensional spectroscopy of electronic couplings in photosynthesis

Tobias Brixner, Jens Stenger, Harsha M. Vaswani, Minhaeng Cho, Robert E. Blankenship & Graham R. Fleming

Nature 434, 625–628(2005) | Cite this article



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Quantum coherence

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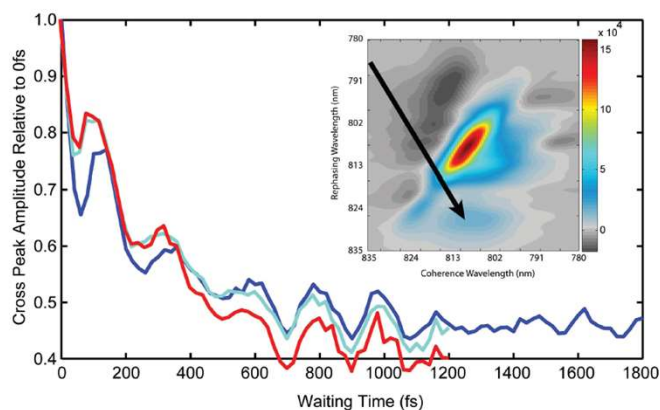
Proceedia Chemistry 3 (2011) 222–231

22nd Solvay Conference on Chemistry

Quantum coherence in photosynthesis

Gregory S. Engel*

The James Franck Institute and Department of Chemistry, The University of Chicago, Chicago, Illinois, USA

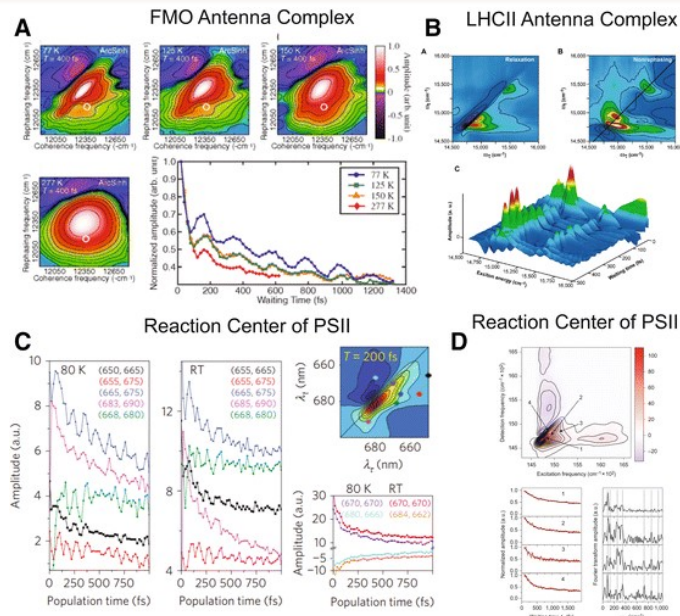


Quantum beating indicating long-lived coherence. Two dimensional spectra of the Fenna-Matthews-Olson complex show clear beating signals in the cross-peak between excitons 1 and 2 as a function of waiting time. The three beating traces represent three replicates prepared and sampled independently.

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Quantum coherence

https://www.researchgate.net/figure/Examples-of-2DES-studies-where-coherent-oscillations-have-been-observed-in-photosynthetic_fig4_323883323



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